

SORTING BERRIES ROBOT ARM CONTROL

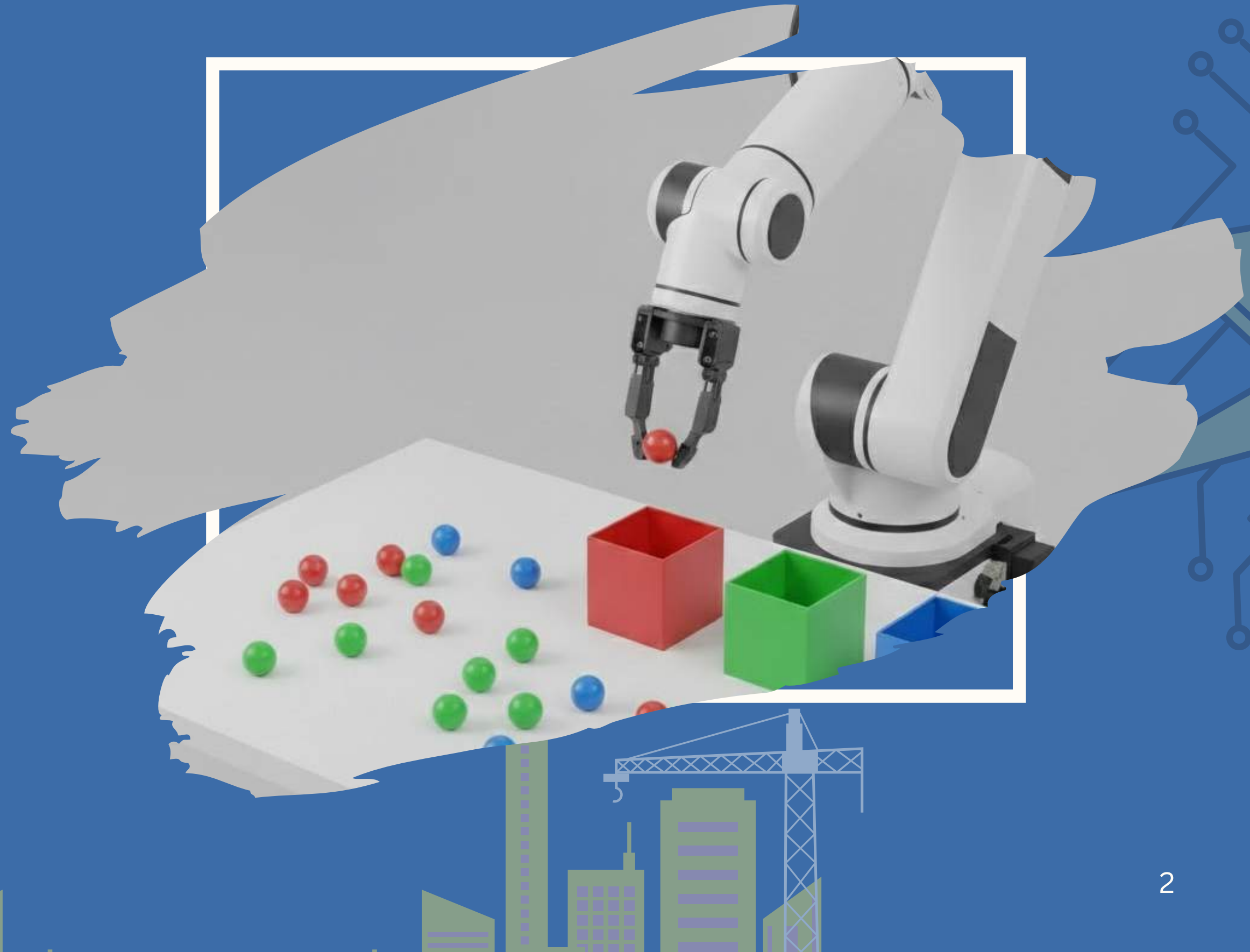
**A Comparison between Resolved Rate of Motion
and Joint Space Trajectory Interpolation**

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MAIN TASK

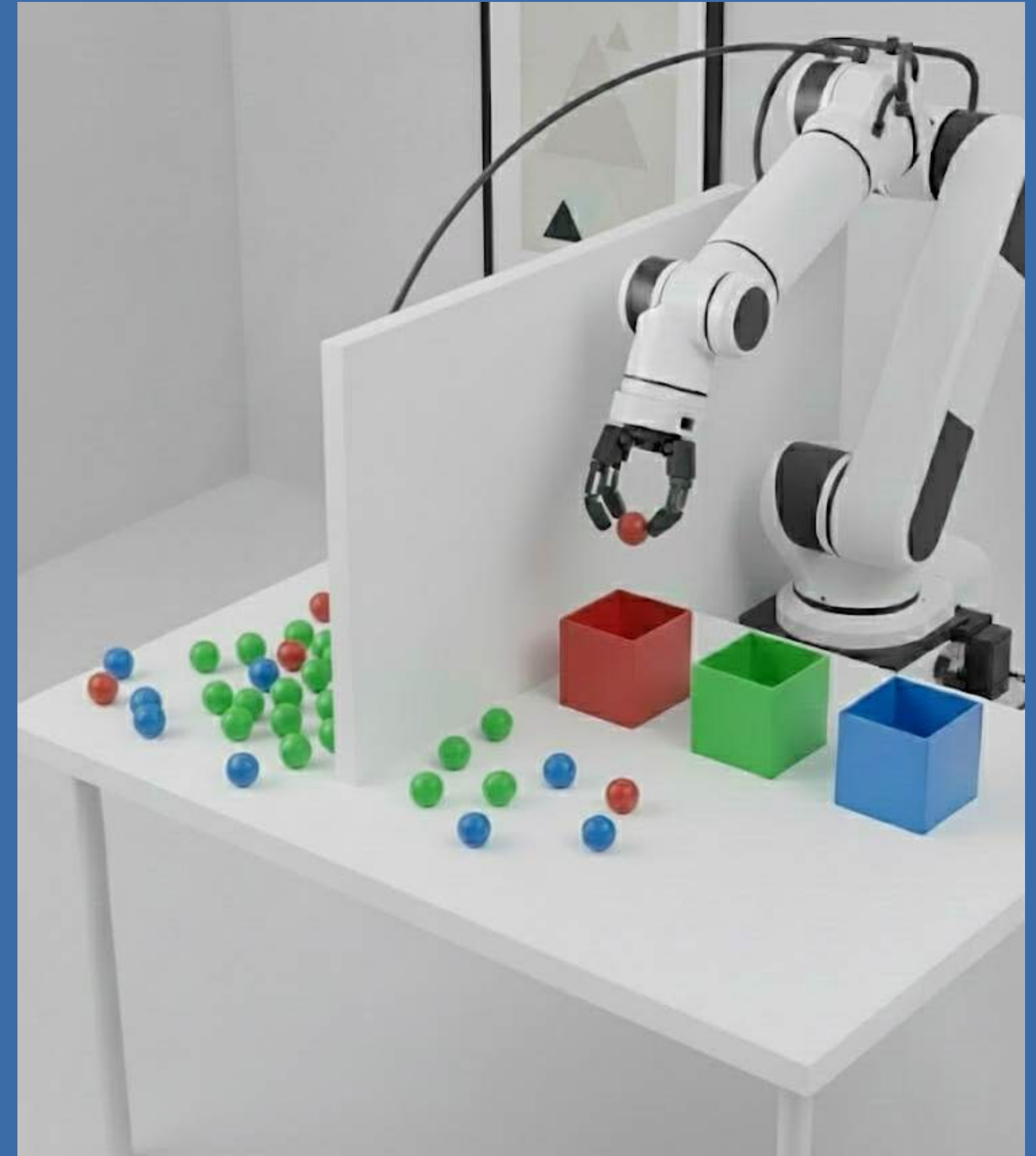
Sorting berries by colour on a table and placing them in the corresponding coloured box using:

1. Resolved Rate Motion Control
2. Joint Space Trajectory Interpolation



MAIN OBJECTIVES

- Comparing the metrics for both strategies for single sorting
- Comparing metrics for both strategy for 100 simulations
- **Extra Steps:** Sorting berries with an obstacle in the trajectory



RESOLVED-RATE OF MOTION (RRMC)

Basic RRMC Equation: $\dot{q} = J^\dagger(q) \cdot K \cdot e$

Where:

- $\dot{q}$: Joint velocities
- $J^\dagger$: Damped pseudo-inverse of the Jacobian matrix
- K : Control gain (1.5 in implementation)
- e : Position error vector

RRMC is a velocity-based control method that uses the robot's Jacobian to continuously calculate the necessary joint velocities to achieve a desired end-effector velocity. This results in smooth, direct Cartesian paths.

JOINT SPACE TRAJECTORY INTERPOLATION

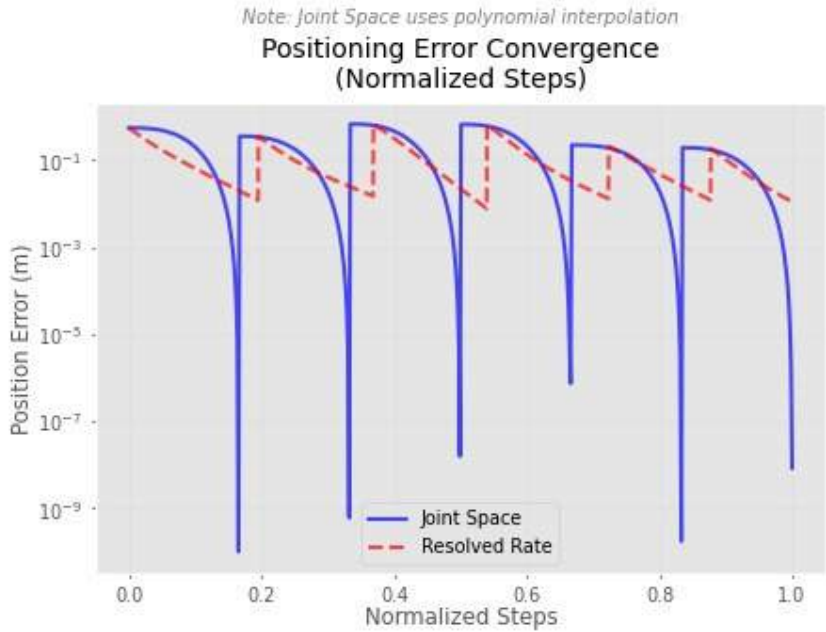
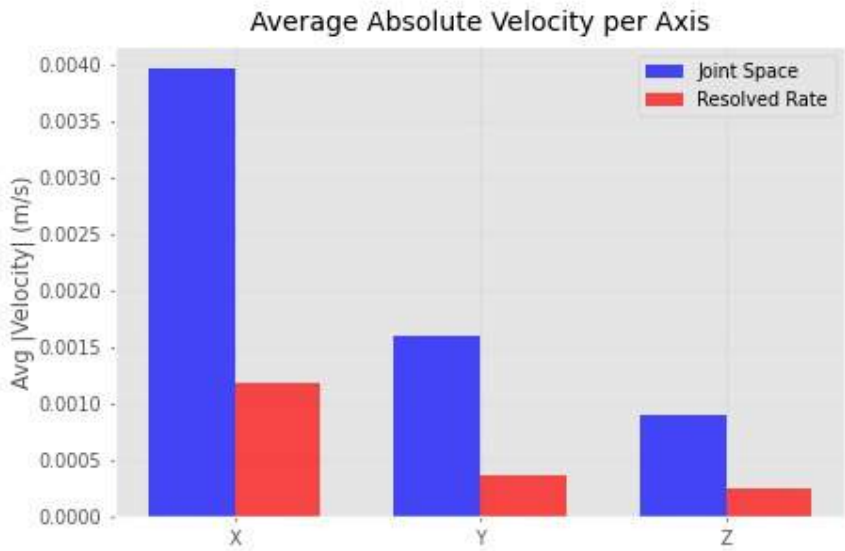
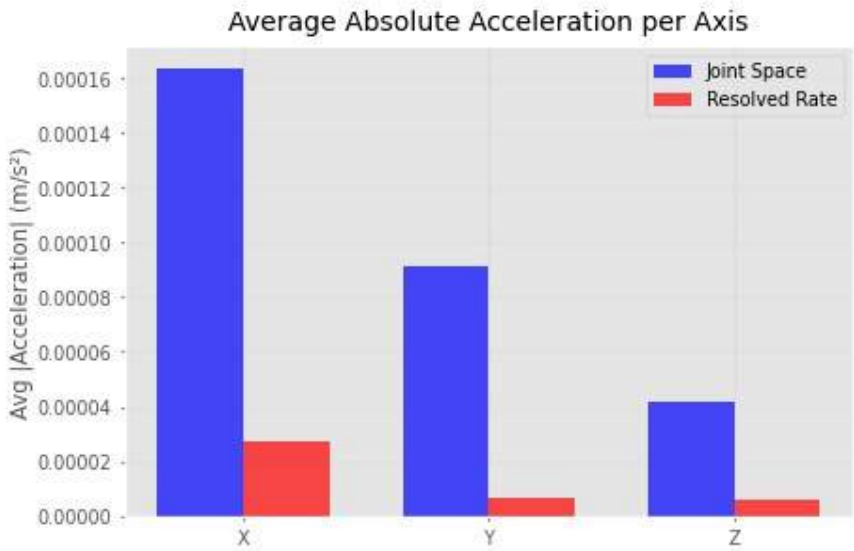
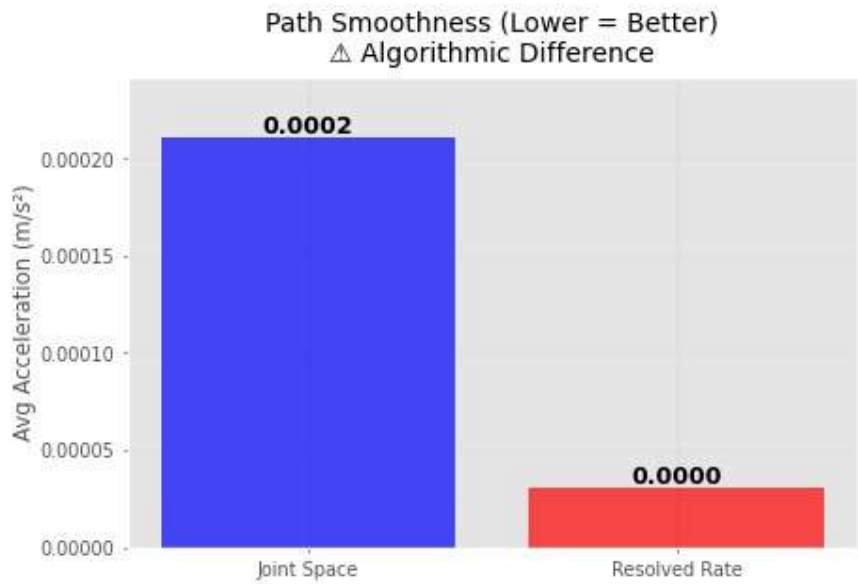
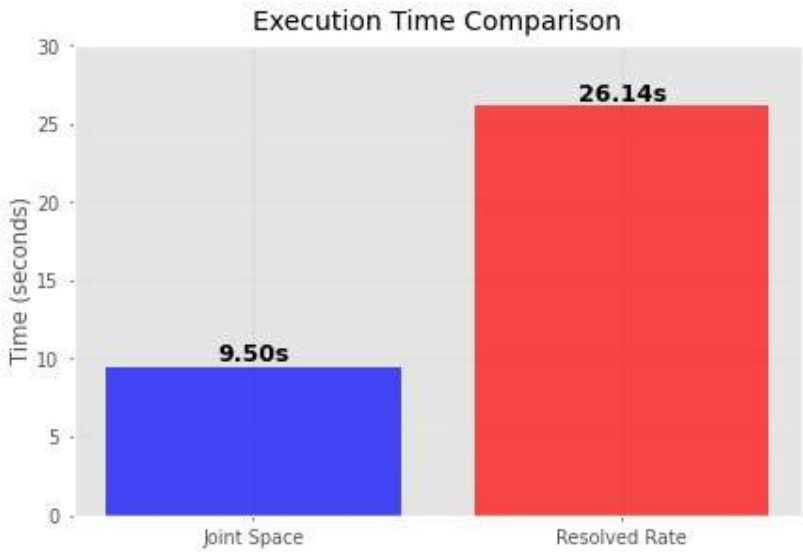
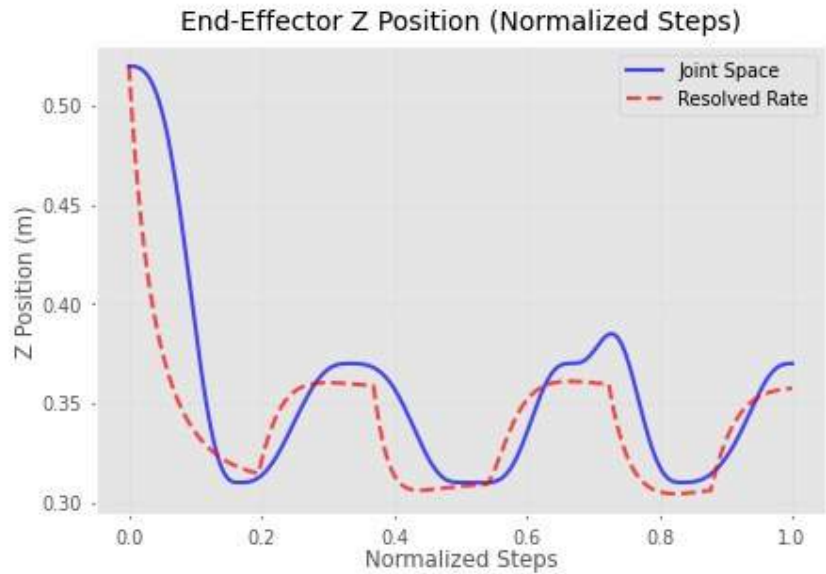
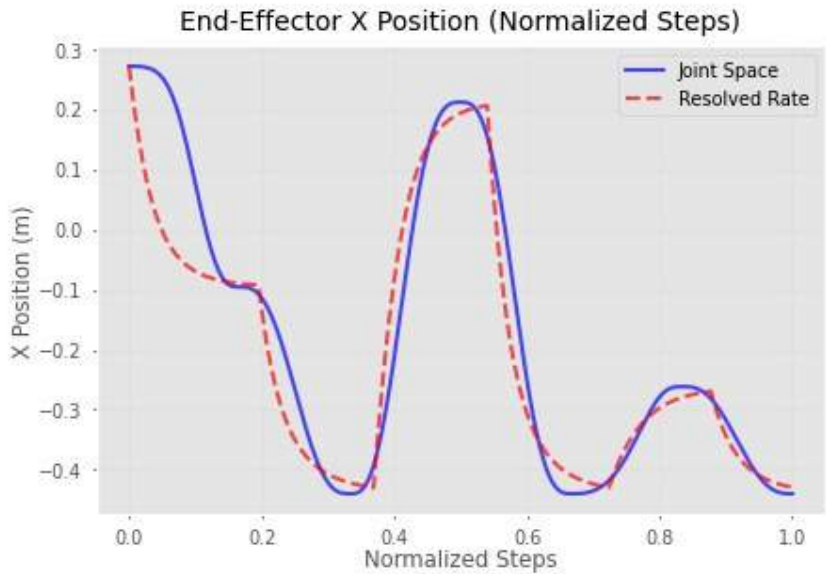
Joint Space Trajectory Interpolation involves planning a trajectory in the robot's joint space.

1. IK solver: Finds joint angles for the start and end poses
2. Interpolation: Fills in all the intermediate joint angles for a smooth trajectory between those two points.
3. The resulting end-effector path in Cartesian space is determined by the joint interpolation.

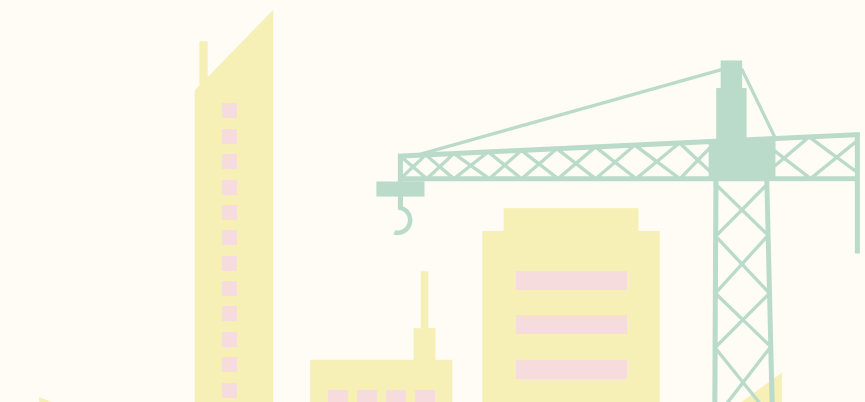
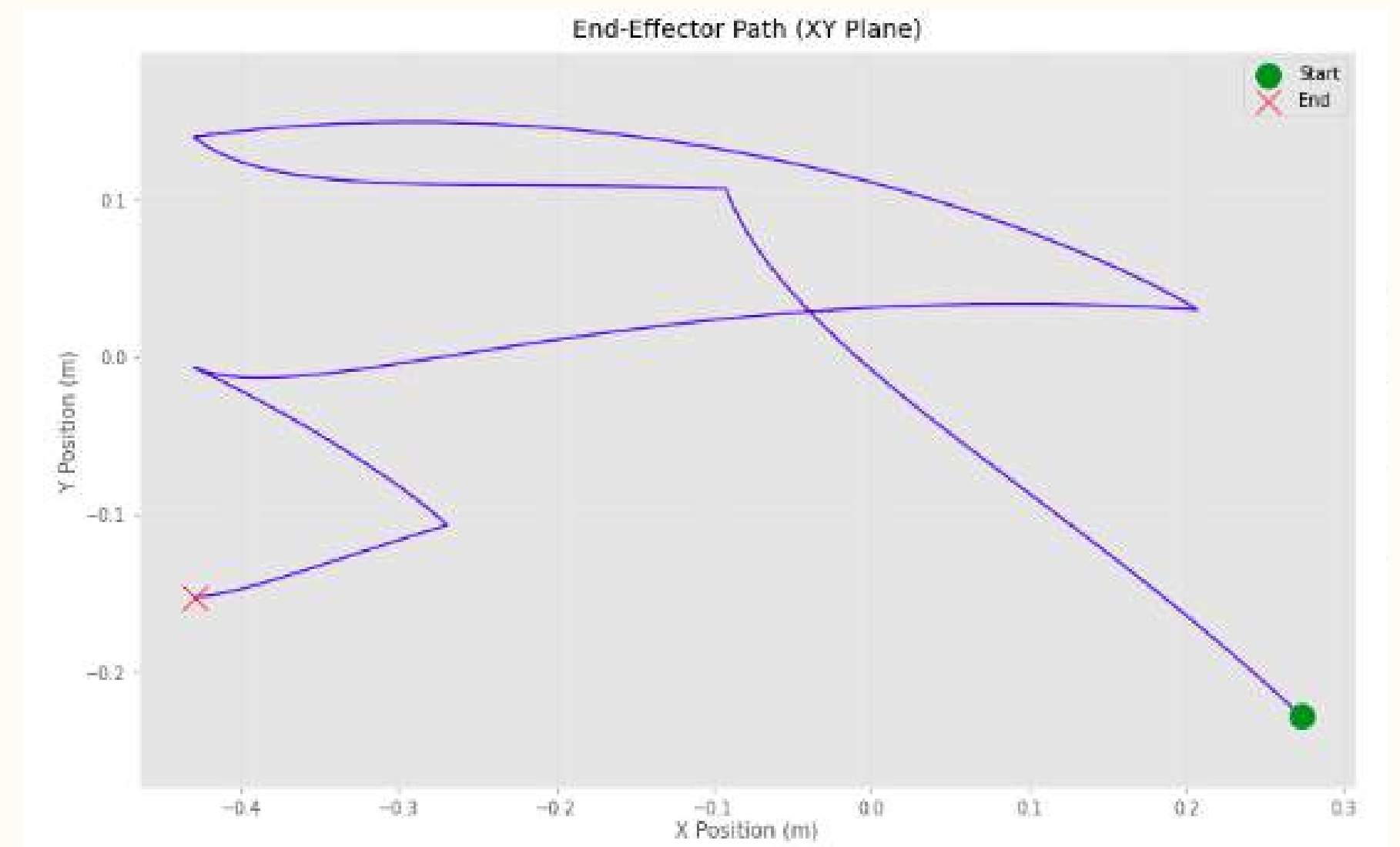
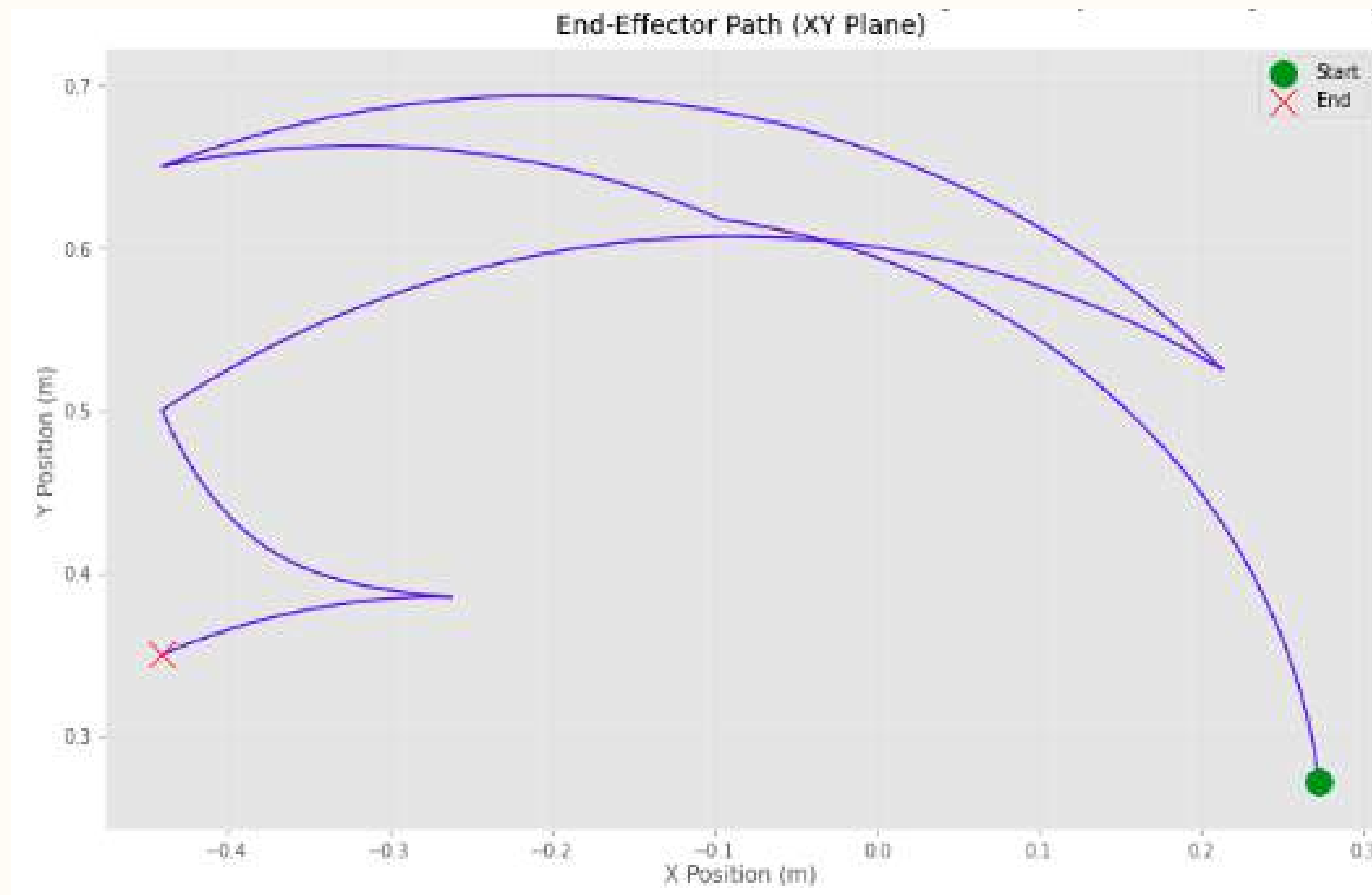
The `rtb.jtraj()` function generates smooth joint space trajectories using 5th-order polynomials:

$$q_i(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

Joint Space vs Resolved-Rate Motion Control: Fair Comparison



Question for the class: Which one is joint space INTERPOLATION and which is resolved IN MOTION?





Simulating the sort 100 times

Resolved Rate Strengths:

- Superior average accuracy (50% better)
- Consistent performance across all runs
- Natural obstacle avoidance integration
- More consistent performance

Joint Space Strengths:

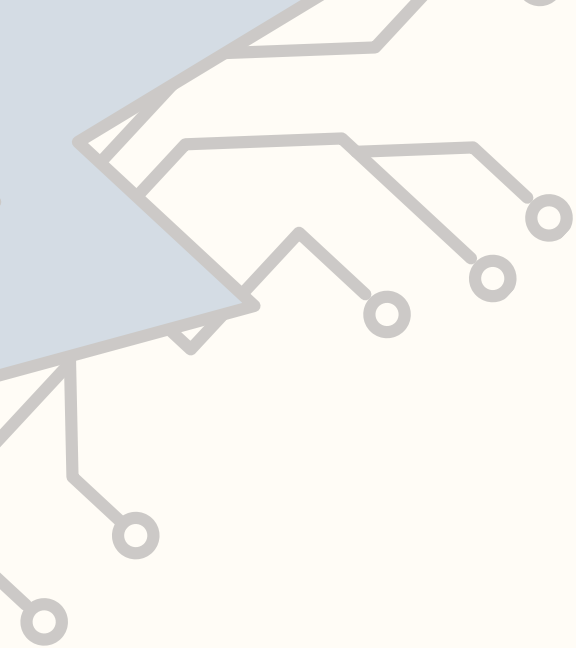
- Perfect final positioning (zero error)
- Shorter path lengths (7% more efficient)
- Predictable, deterministic behavior
- Guaranteed smooth trajectories (polynomial interpolation)

Performance Metrics Comparison

Metric	Resolved Rate	Joint Space	Winner	Improvement
Realistic Task Time (s)	58.91 ± 3.08	35.52 ± 4.57	Joint Space	65.9% faster
Average Error (m)	0.1134 ± 0.0102	0.2289 ± 0.0323	Resolved Rate	50.4% better
Final Error (m)	0.0128 ± 0.0023	0.0000 ± 0.0000	Joint Space	Perfect accuracy
Maximum Error (m)	0.7653 ± 0.0534	0.7529 ± 0.0722	Joint Space	1.6% better
Path Length (m)	9.0557 ± 1.1978	8.4739 ± 1.6366	Joint Space	6.9% shorter
Mean Velocity (m/s)	11.62 ± 1.33	6.20 ± 1.94	Resolved Rate	87.5% faster

Algorithmic-Dependent Metrics

Metric	Resolved Rate	Joint Space	Notes
Path Smoothness (m/s²)	0.0000 ± 0.0000	0.0002 ± 0.0000	<i>Reflects control method</i>
Total Steps	6538 ± 281	1776 ± 229	<i>Joint Space: fixed trajectory</i>



Real World Selection Guidelines

Choose Resolved Rate for:

- Tasks requiring adaptive, real-time control
- Applications prioritizing motion accuracy
- Environments with dynamic obstacles

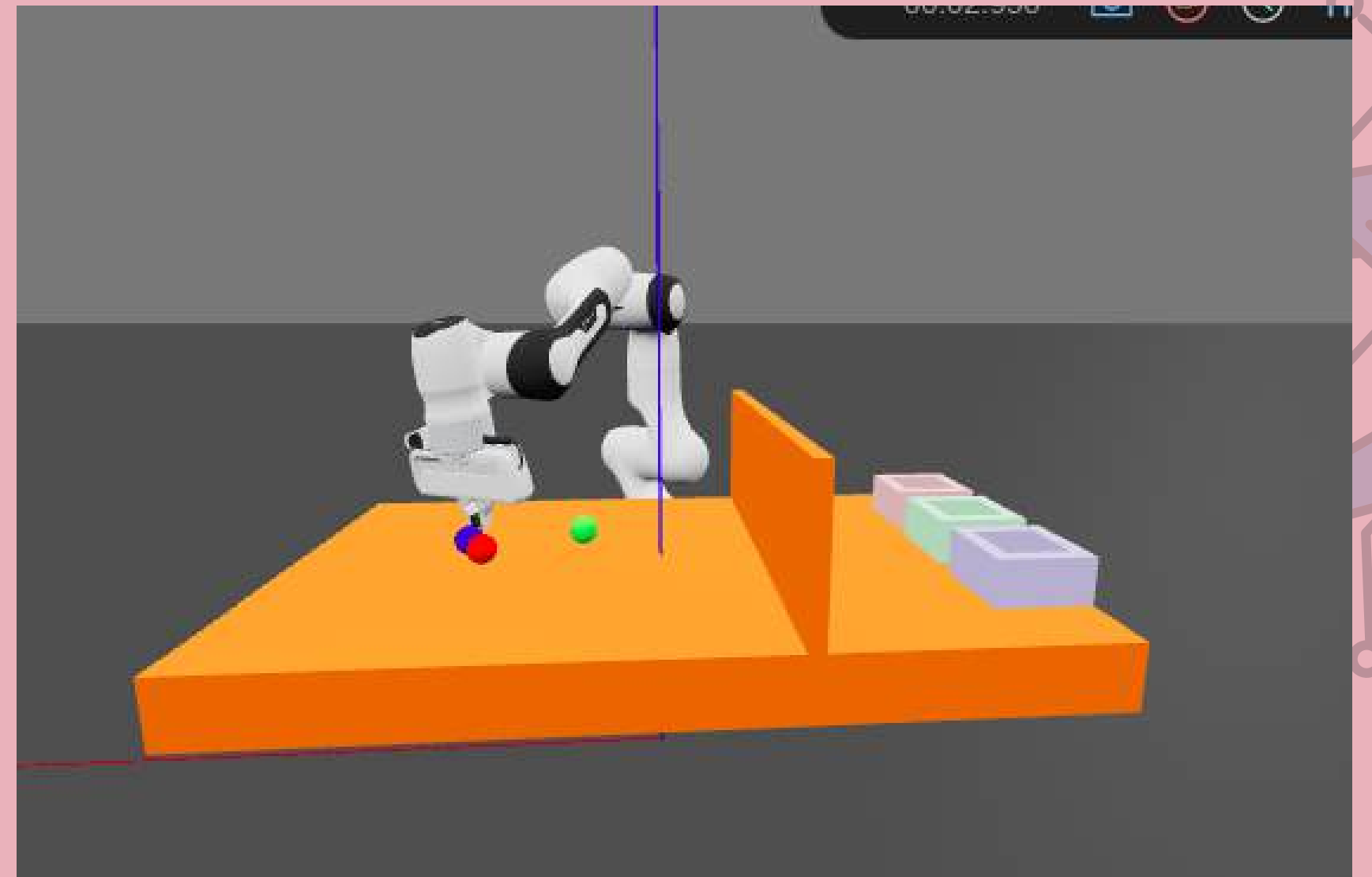
Choose Joint Space for:

- Production systems needing predictable timing
- Safety-critical or deterministic applications
- Scenarios prioritizing task completion speed

OBJECT AVOIDANCE

Our technique:

- target = natural attractive force
- object = repulsive force
- cubic repulsive force that increases exponentially as the end-effector gets closer to the obstacle



REPULSIVE FORCE FORMULA

Repulsive Force (from obstacles):

$$F_{repulsive} = \sum_i F_{rep,i} + F_{upward,i}$$

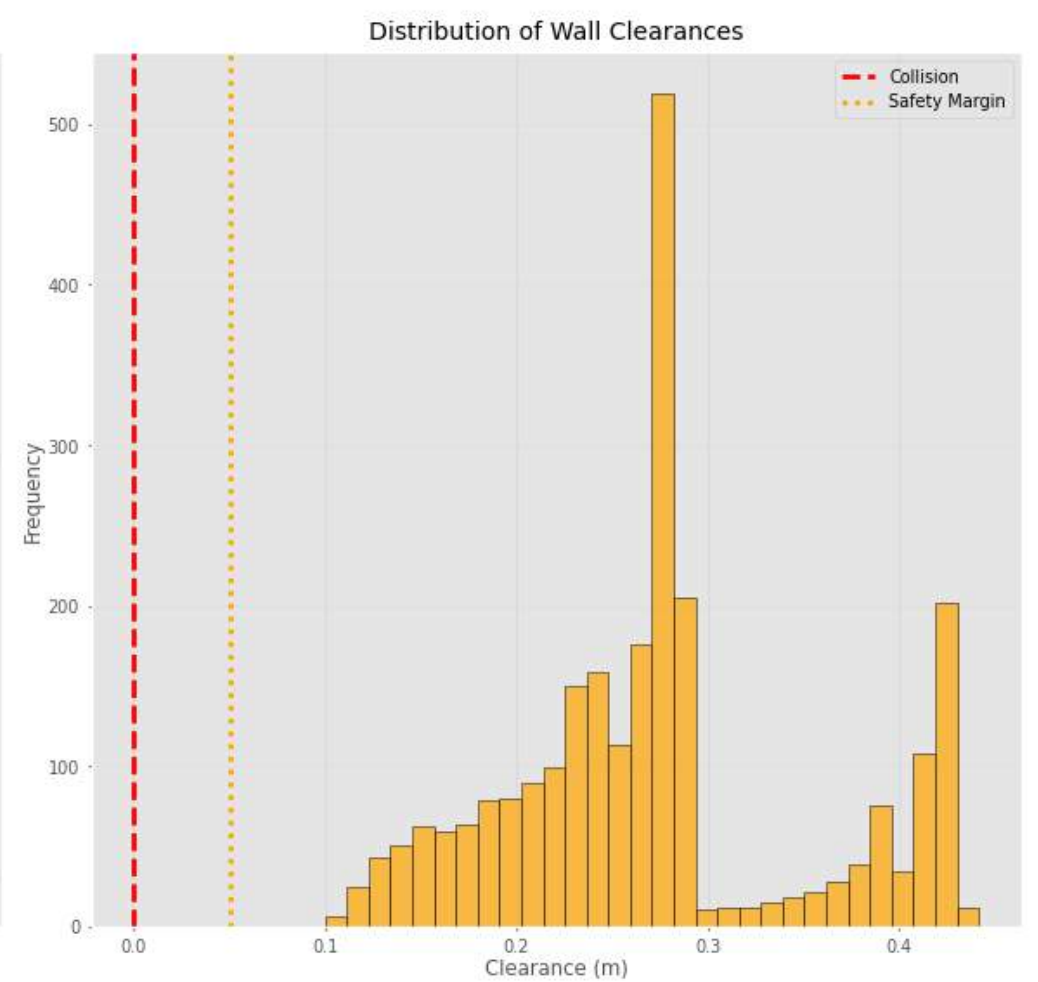
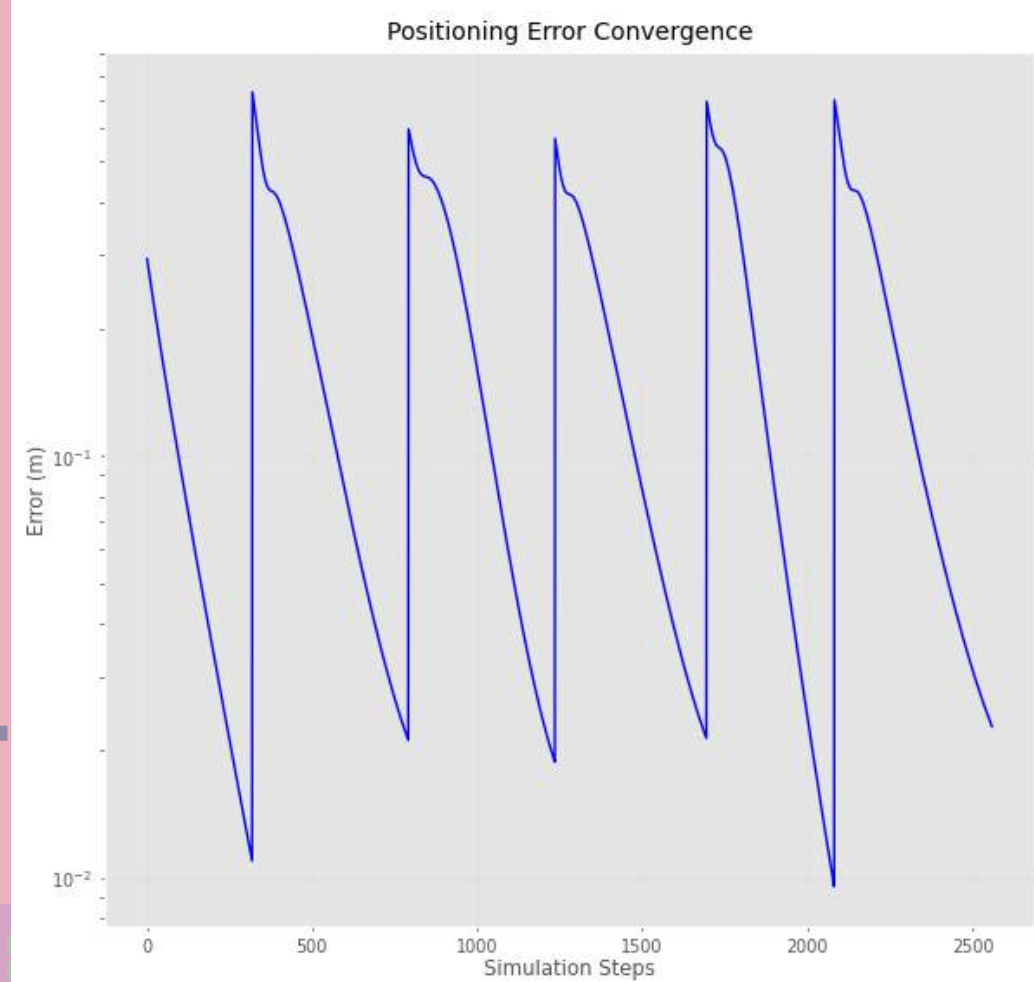
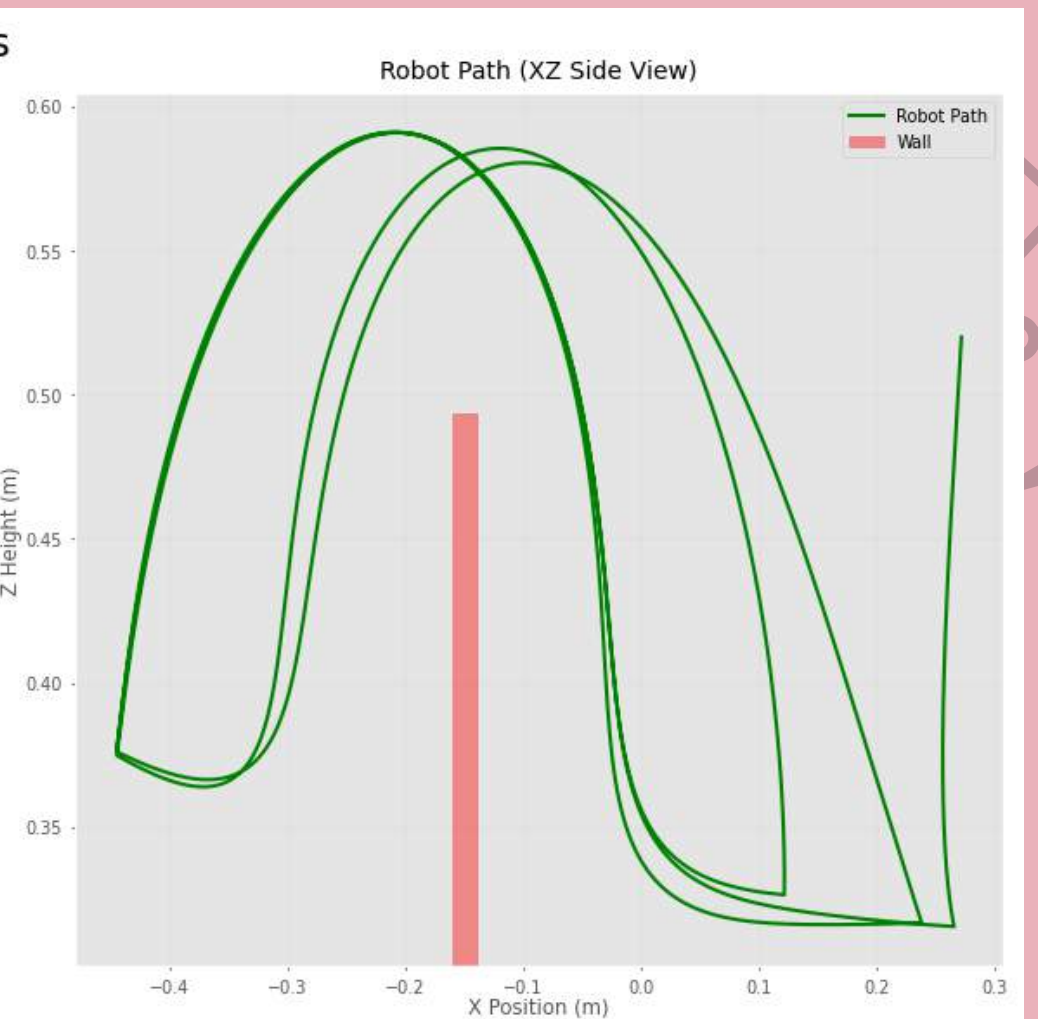
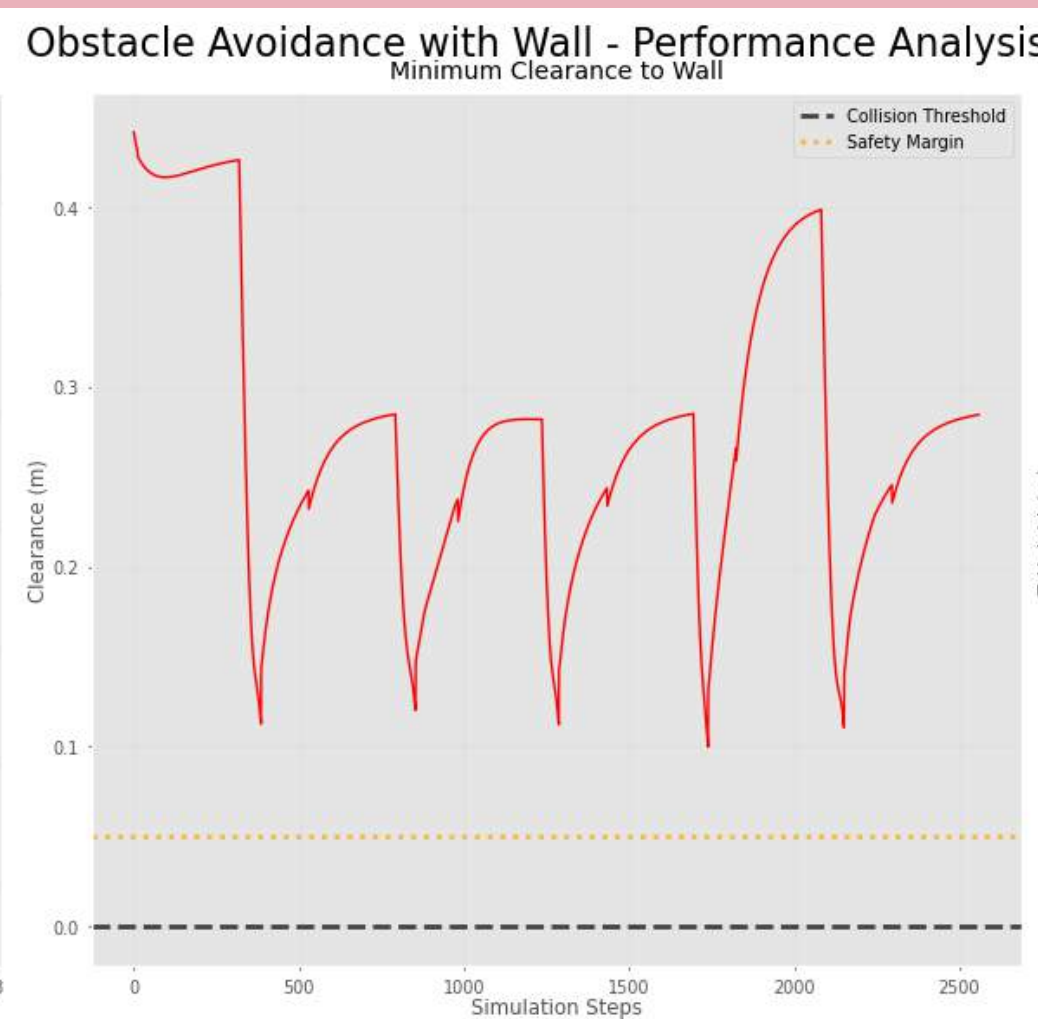
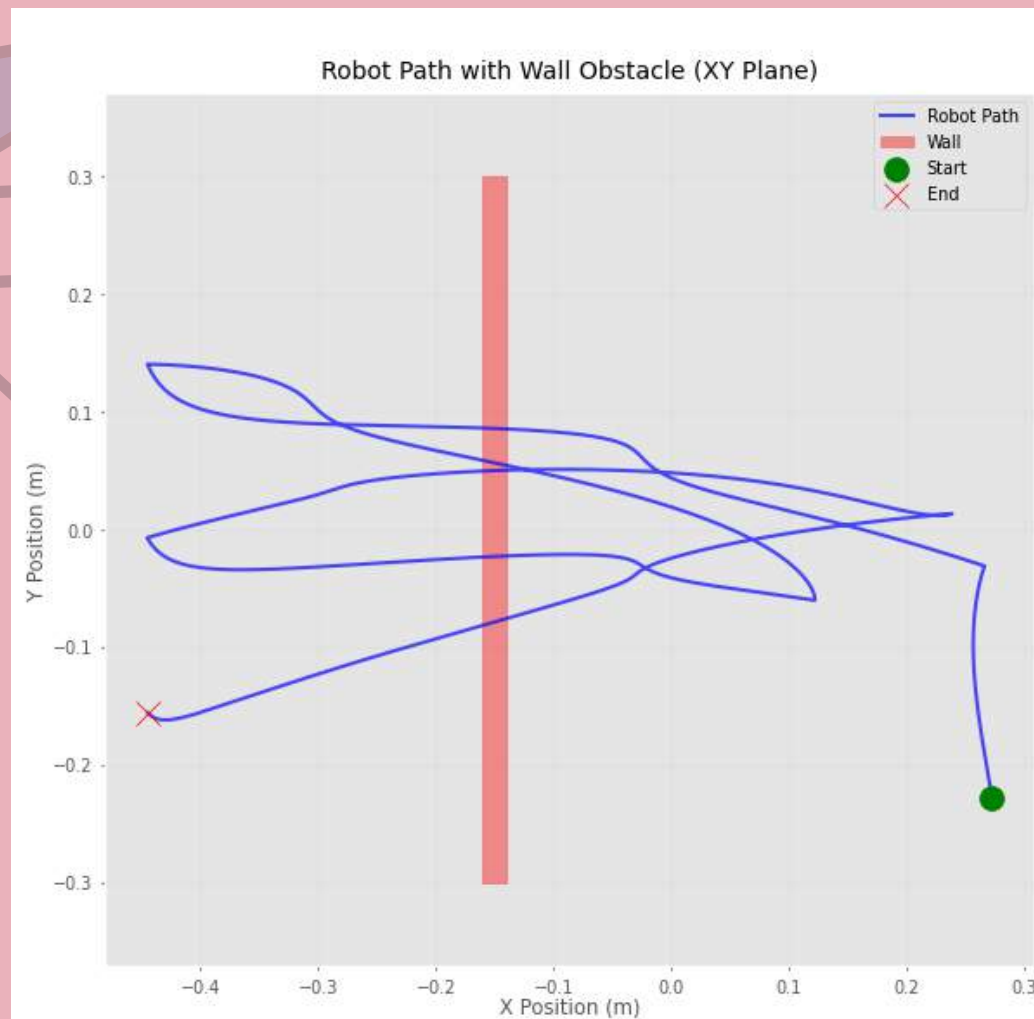
For each obstacle i :

$$F_{rep,i} = K_{rep} \cdot \left(1 - \frac{d_i}{d_{influence}}\right)^3 \cdot \hat{n}_i$$

$$F_{upward,i} = [0, 0, K_{rep} \cdot \left(1 - \frac{d_i}{d_{influence}}\right)^3]$$

Where:

- K_{rep} : Repulsive gain (0.1)
- d_i : Distance to obstacle i surface
- $d_{influence}$: Influence distance (0.30 m)
- $\hat{n}_i$: Unit vector pointing away from obstacle
- $F_{upward,i}$: Additional upward force to encourage going over obstacles



Execution Statistics:

Total Time: 29.23 s
Total Steps: 2560

Wall Avoidance:
Min Clearance: 0.1000 m
Avg Clearance: 0.2737 m
Safety Violations: 0 steps
(0.0% of path)
Collisions: N0

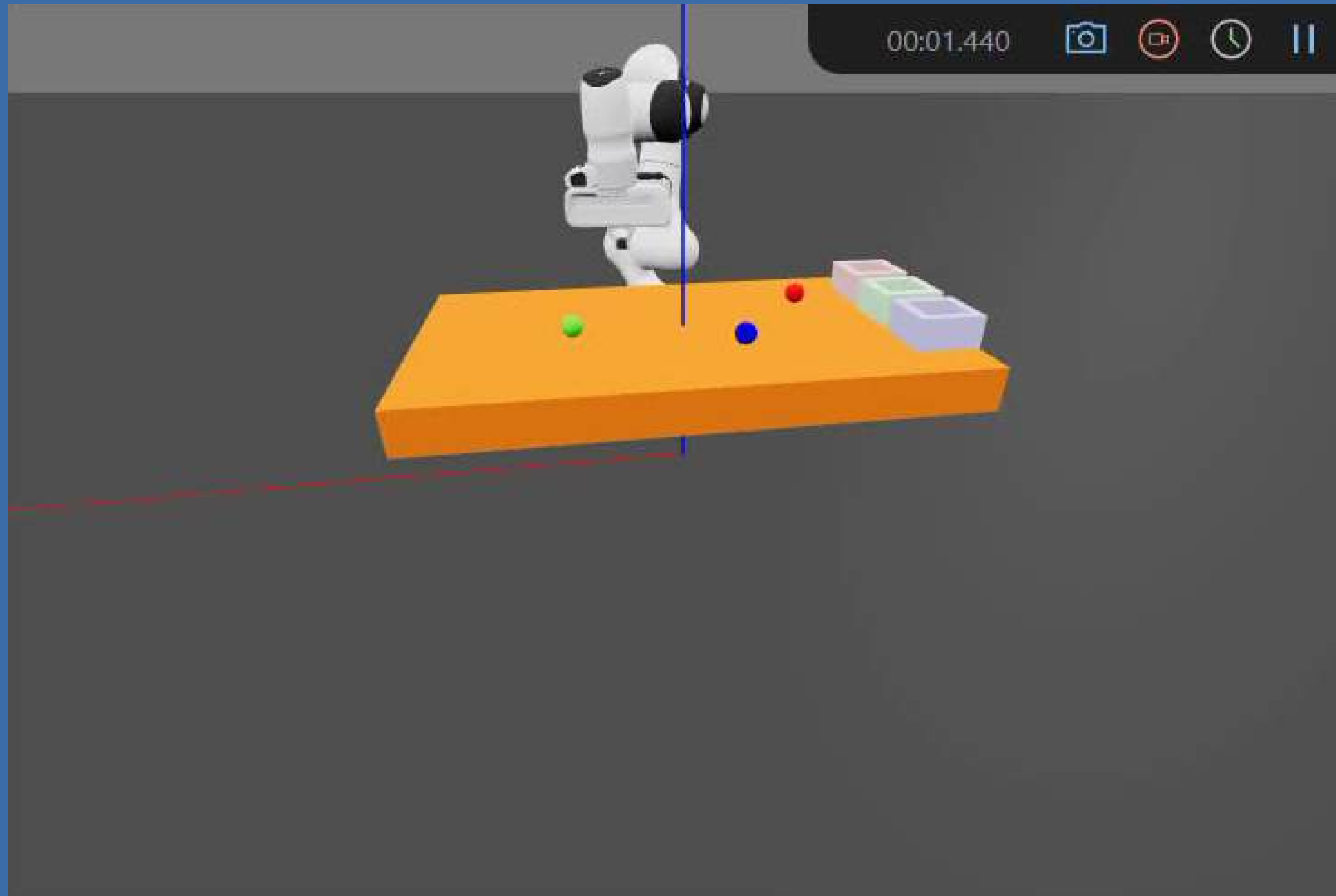
Positioning:
Avg Error: 0.177825 m
Final Error: 0.022847 m

Path Efficiency:
Total Path Points: 2560

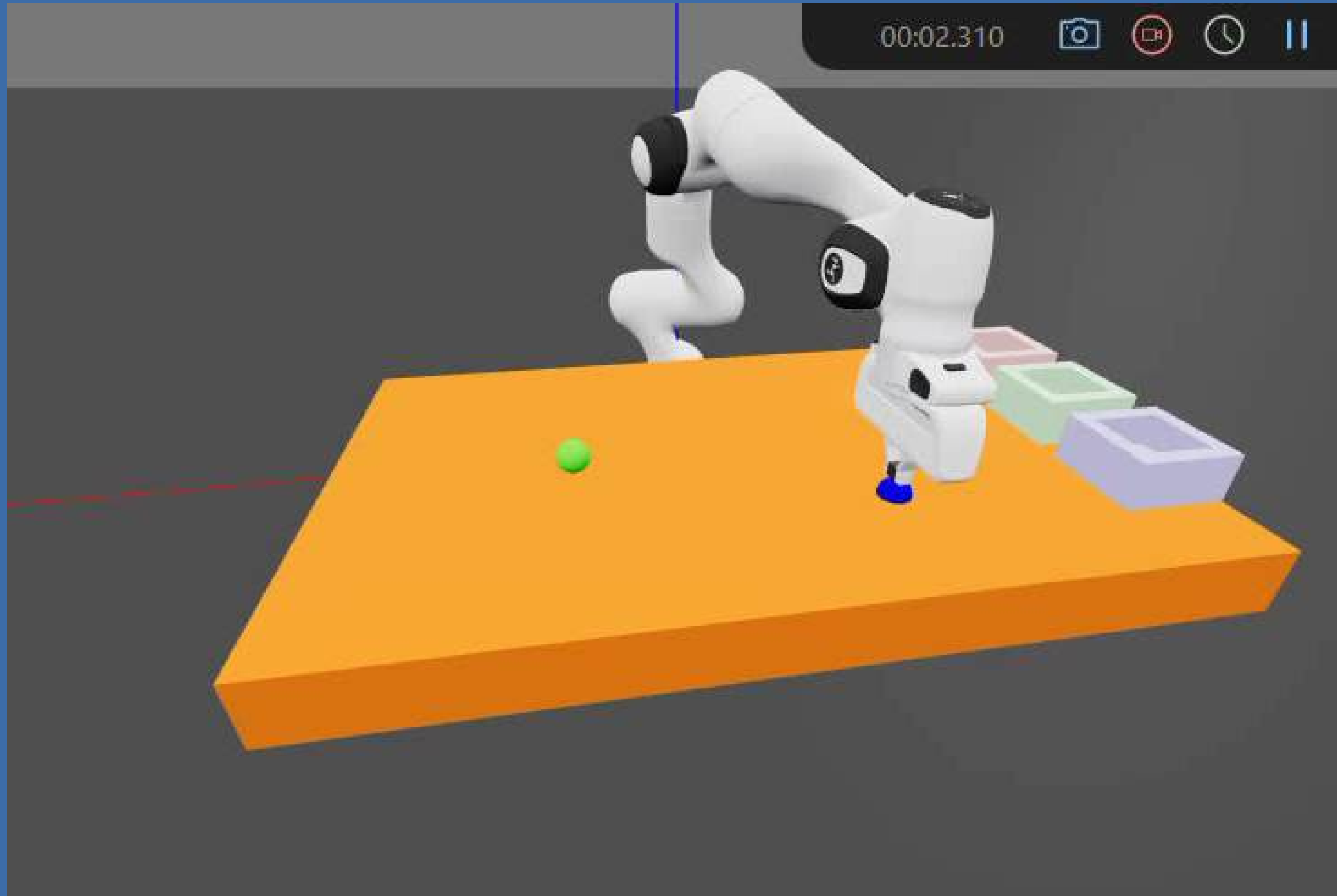


SHOWCASE!

RESOLVED RATE OF MOTION



JOINT SPACE TRAJECTORY INTERPOLATION



RRM WITH OBSTACLE

